

Data Structure

Level 1

Singly Linked List Data Structures Operations

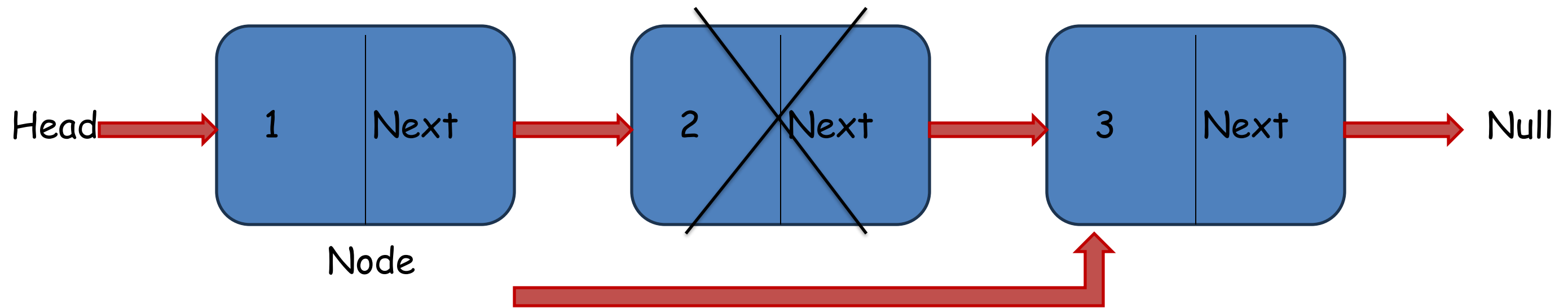
The background of the slide features a complex, abstract network of thin, dark lines crisscrossing a light gray background, resembling a web or a neural network. In the lower portion of the image, there are dark silhouettes of several people, some standing and some leaning over, suggesting a collaborative work environment or a meeting. A large, dark gray rectangular area is overlaid on the right side of the image, serving as a backdrop for the text.

Data Structure

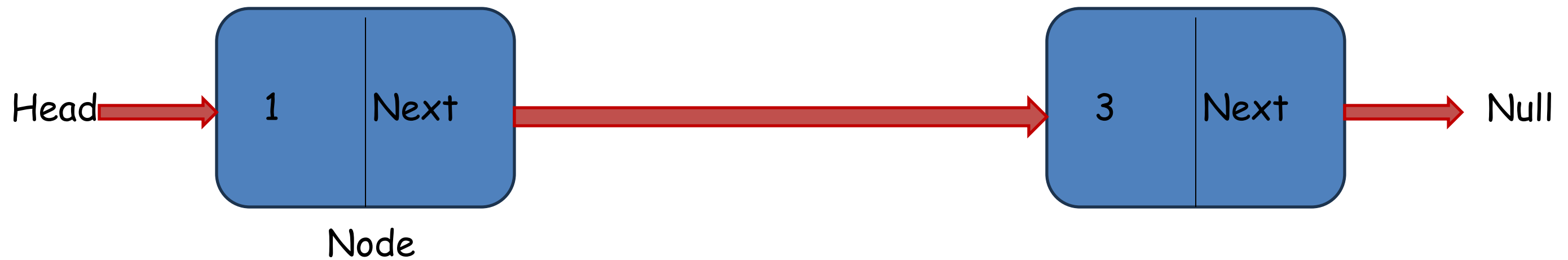
Level 1

Delete Node

Linked List Data Structure



Linked List Data Structure



Definition

DeleteNode() Function:

Removes the first node that contains a given value from the linked list and frees its memory.

 Responsibilities:

-  Locate the node to delete
-  Update links between nodes
-  Free memory using delete



Explanation

- ◆ If the list is empty → nothing to delete.
- ◆ If the node to delete is the head, move head to the next node.
- ◆ Otherwise, traverse the list while keeping track of:
 - Current → node being checked
 - Prev → previous node
- ◆ Once found, connect Prev->next to Current->next.
- ◆ Free memory using delete Current.

Procedures (Delete Node)

- 1 Check if list is empty
- 2 Set Current = head and Prev = head
- 3 If head contains the value:
 -  Move head to Current->next
 -  Delete Current
- 4 Otherwise, traverse list
- 5 Move Prev and Current forward
- 6 Stop when value is found
- 7 If value not found → return
- 8 Update Prev->next = Current->next
- 9 Delete Current from memory

Example

Initial operations in main():

InsertAtEnd(1)

InsertAtEnd(2)

InsertAtEnd(3)

InsertAtBeginning(0)

→ List becomes:

0 → 1 → 2 → 3 → NULL

DeleteNode(head, 4)

→ Since 4 is not in the list → no change

Final list remains:

0 → 1 → 2 → 3 → NULL

Note

The function deletes only the first occurrence of the value.



Comparison

- Delete Head Node
 - ✓ $O(1)$ time
 - ✓ Simple pointer update

- Delete Middle/Last Node
 - ✓ Requires traversal
 - ✓ Time Complexity: $O(n)$

Warning

If you forget delete Current, it causes a memory leak.

If you update pointers incorrectly, the list structure may break.

Summary

- ✓ DeleteNode removes node by value
- ✓ Handles head deletion separately
- ✓ Requires traversal for non-head nodes
- ✓ Time complexity is $O(n)$
- ✓ Memory must be freed using delete

Thank you 😊